

Lecture 2

Microeconomic Theory III – ECON 709
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Recap' Lecture 1

- ▶ A preference relation $\succsim$ on a set X is *rational* if it satisfies the following three properties:
 1. *Reflexivity*: For any $x \in X$, $x \succsim x$.
 2. *Completeness*: For any distinct $x, y \in X$, $x \succsim y$ or $y \succsim x$.
 3. *Transitivity*: For any $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.
- ▶ If $\succsim$ is rational then:
 1. $\succ$ is irreflexive and transitive.
 2. $\sim$ is reflexive, transitive and symmetric.
 3. If $x \succ y \succsim z$, then $x \succ z$.
- ▶ Are the “rationality” properties of $\succsim$ sensible assumptions?

Reflexivity: For any $x \in X$, $x \succsim x$.

- ▶ means that when comparing an alternative with itself, the alternative is at least as good as itself
- ▶ implies indifference between two identical options, i.e. $x \sim x$

Completeness: For any distinct $x, y \in X$, $x \succsim y$ or $y \succsim x$.

- ▶ completeness implies that the preference relation is well-defined for any two distinct alternatives; that is, for any pair of distinct x and y in X , we have one of the three mutually exclusive possibilities:
 - ▶ $x \succsim y$ and not $y \succsim x \Leftrightarrow x \succ y$ (x is strictly better than y)
 - ▶ $y \succsim x$ and not $x \succsim y \Leftrightarrow y \succ x$ (y is strictly better than x)
 - ▶ $x \succsim y$ and $y \succsim x \Leftrightarrow x \sim y$ (x is as good as y)
- ▶ any two distinct alternatives x and y can be ranked
- ▶ this is a strong assumption:
 - ▶ requires you to always know your preference
 - ▶ even though alternatives might not be well known to you
- ▶ but useful: how could choice/satisfaction over alternatives be analyzed if preferences were undefined?

Transitivity: For any $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

Examples:

- ▶ If "large" is preferred over "medium", and "medium" over "small", then "large" is also preferred over "small". This seems plausible.
- ▶ If "red" is preferred over "blue" and "blue" over "yellow", then "red" should also be preferred over "yellow". This is less obvious.

Transitivity is a strong assumption. We can easily think of plausible intransitive preference relations...

Potential Sources of Intransitivity

1. Deriving a preference relation over alternatives by aggregating preference relations over components:

- ▶ suppose alternatives in X consist of several components and an individual derives her preference relation $\succsim$ over X by aggregating her preferences relations for the different components, for example
" $x \succsim y$ iff a majority of the components in x is preferred over the components in y "
- ▶ e.g.:
 - ▶ set of holiday packages $X = \{a, b, c\}$ and each $i \in X$ is characterized by its price p_i , location l_i , and food f_i
 - ▶ assume the following preference relations over prices, locations, and food:
 $p_a \succ_p p_b \succ_p p_c$
 $l_b \succ_l l_c \succ_l l_a$
 $f_c \succ_f f_a \succ_f f_b$
 - ▶ if above rule is applied, then $a \succ b$, $b \succ c$, and $c \succ a$, and thus the preference relation over X violates transitivity

Potential Sources of Intransitivity

2. Use of similarities to establish indifference:

- ▶ suppose an individual expresses indifference between two alternatives that are too "close" to be distinguishable for her
- ▶ example:
 - ▶ let X be the set of real numbers
 - ▶ consider an individual whose attitude toward the alternatives is "the larger the better" however she cannot determine whether $x > y$ unless the difference is at least 1
 - ▶ thus $x \succ y$ if $x \geq y + 1$ and $x \sim y$ if $|x - y| < 1$
 - ▶ then $1.5 \sim .8$ and $.8 \sim .3$ but $1.5 \succ .3$ so it is not true that $1.5 \sim .3$, violating transitivity

Then why do we assume transitivity?

- ▶ We avoid "cycles" in preferences, like $a \succ b$, $b \succ c$, and $c \succ a$.
- ▶ With cycles, there exists a sequence of pairwise choices such that the decision maker always prefers the "newly" offered alternative.
- ▶ With cycles, we might not be able to determine the set of "best" alternatives from X . Determining a "best" alternative is important for choice analysis.
- ▶ And in many situations, transitivity is a natural property of preferences...

Implicit Assumptions on Preference Relations

- ▶ no intensities of preferences:
a preference relation cannot capture the difference between "I somewhat prefer x over y " vs. "I love x and hate y "
- ▶ no dependency on other factors:
e.g. it is impossible to, depending on circumstances, sometimes prefer x over y , and sometimes y over x

Equivalent Definitions of Rational Preference Relations

1. combining reflexivity and completeness, e.g. MWG:
 - ▶ "A preference relation is rational if it satisfies 1) Transitivity, and 2) Completeness: 'For any $x, y \in X$, $x \succsim y$ or $y \succsim x$ '."
 - ▶ The definition of completeness is extended and includes reflexivity. Different ways of saying one thing!
2. avoiding the terminology "rational" (e.g. Rubinstein):
 - ▶ "A binary relation $\succsim$ is a preference relation if $\succsim$ is 1) reflexive (R), 2) complete (C), and 3) transitive (T)."
 - ▶ according to this definition, every preference relation is "rational" (R, C, T), and if a binary relation $\succsim$ is not "rational", then it is not a preference relation
 - ▶ this avoids classifying R, C, and T as "rational" properties
 - ▶ terminology "rational" is debated because you can think of "sensible" tastes/opinions not satisfying R, C, T
 - ▶ it gives up calling a binary relation $\succsim$ a preference relation if $\succsim$ is not R, C, T
 - ▶ in the end, only a different terminology is used here

We'll stick to Definition 1...

Utility

based on: Rubinstein, lecture 2, lecture 4 (proof of Debreu's Theorem); MWG, sections 1.B, 3.C

The concept of utility representation

- ▶ often the natural description of a preference relation involves a numerical representation, e.g.:
 - ▶ “I prefer the taller basketball player”, “I prefer the more expensive present”, “I prefer the smaller car”
- ▶ such a preference relation can be naturally specified by a statement of the form “ $x \succsim y$ if $V(x) \geq V(y)$ ” (or $V(x) \leq V(y)$), where $V : X \rightarrow \mathbb{R}$ is a function that attaches a real number to each element in the set of alternatives X
- ▶ e.g. the preferences stated by “I prefer the taller basketball player” can be expressed formally by:
 X is the set of all conceivable basketball players, and $V(x)$ is the height of player x .
- ▶ the statement $x \succsim y$ if $V(x) \geq V(y)$ always defines a preference relation because the relation $\geq$ on $\mathbb{R}$ satisfies completeness and transitivity

Even when the natural description of a preference relation does not involve a numerical evaluation, we are interested in an equivalent numerical representation...

Utility functions!

Utility Functions

- ▶ we often describe a preference relation by a *utility function*
- ▶ a utility function $u(x)$ assigns a numerical value to each element in X , ranking the elements of X in accordance with the preference relation, more precisely:

Definition 2

A function $u : X \rightarrow \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$, $x \succsim y \Leftrightarrow u(x) \geq u(y)$.

- ▶ A utility function representing preference relation $\succsim$ (utility representation of $\succsim$) merely summarizes the information contained in $\succsim$, by assigning higher numbers to higher ranked bundles (not more, not less).
- ▶ a utility representation of $\succsim$ is useful: e.g. use mathematical programming to determine most and least preferred alternatives (function maximization and minimization)

Non-Uniqueness of Utility Representations

If a preference relation $\succsim$ can be represented by a utility function, then there exist infinitely many such utility functions, namely, all functions which “rank” the alternatives in the same way as $\succsim$ are a utility representation of $\succsim$.

Proposition 2

If u represents $\succsim$, then for any strictly increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$, the function $v(x) = f(u(x))$ represents $\succsim$ as well.

Proof.

$$x \succsim y$$

$$\Leftrightarrow u(x) \geq u(y) \text{ (since } u \text{ represents } \succsim)$$

$$\Leftrightarrow f(u(x)) \geq f(u(y)) \text{ (since } f \text{ is strictly increasing)}$$

$$\Leftrightarrow v(x) \geq v(y)$$



Ordinal and cardinal properties of a utility function

- ▶ The representation of a preference relation $\succsim$ by a utility function is an *ordinal* property of the utility function. For the representation of $\succsim$, we only care about the *ordering* of alternatives reproduced by the utility function, not about the absolute numbers attached to alternatives. Ordinal properties are preserved for all transformations mentioned in Prop. 2.
- ▶ The actual functional form and thus numerical values attached to alternatives are *cardinal* properties of a utility function. Cardinal properties are not preserved for transformations mentioned in Prop. 2. If we construct a utility function in order to define a preference relation, then cardinal properties usually carry an additional, intuitive meaning about the preferences. E.g. the magnitude of differences between utility values might imply something about the intensity of preferences (a is slightly preferred over b , while c is "extremely" preferred over d).

Existence of a Utility Representation for $\succsim$ on X

- ▶ Can we map all the alternatives in X onto one line such that higher ranked bundles get assigned "higher" positions on the line? i.e. can we assign a real number (utility value) to each alternative in X in an order-preserving way?
- ▶ we will see that not every preference relation can be represented/described by a utility function, not even every rational one
- ▶ thus we cannot just use utility functions and forget about preference relations "without loss of generality"
- ▶ in the following, we present conditions on the preference relation and on X such that the preference relation on X can be described by a utility function

Rationality of $\succsim$ is necessary for a Utility Representation

Proposition 3

A preference relation $\succsim$ can be represented by a utility function, only if it is rational.

Proof.

Show that if $\succsim$ has a utility representation, then $\succsim$ must be rational.

Reflexivity and Completeness. Since u is a real-valued function defined on X , it must be that for any $x \in X$, $u(x) \geq u(x)$, and for any distinct $x, y \in X$, $u(x) \geq u(y)$ or $u(y) \geq u(x)$. As u represents $\succsim$, this implies that $x \succsim x$, and $x \succsim y$ or $y \succsim x$. Thus $\succsim$ is reflexive and complete.

Transitivity. Suppose that $x \succsim y$ and $y \succsim z$. As u represents $\succsim$, $u(x) \geq u(y)$ and $u(y) \geq u(z)$. Therefore, $u(x) \geq u(z)$ and since u represents $\succsim$, this implies $x \succsim z$. Thus $\succsim$ is transitive. $\square$

So does any rational preference relation have a utility representation?

No!

We will next see under which assumptions a rational preference relation has a utility representation...

If X is finite, then a rational $\succsim$ has a utility representation.

- ▶ a finite set is a set which has a finite number of elements; informally, one can count the elements and finish counting
- ▶ Intuition for why the above claim is true? Assigning utility values (from the real line with infinitely many numbers) in an order-preserving way to a finite set of ordered alternatives of course works!
- ▶ we present a very detailed proof of this claim, mostly to get into the habit of analytical precision
- ▶ the proof starts with a lemma regarding the existence of minimal elements
(an element $a \in X$ is *minimal* if $x \succsim a$ for any $x \in X$).

Lemma 1

If $\succsim$ is rational, then, in any finite set $A \subseteq X$, there is a minimal element (similarly, there is a maximal element).

Proof.

- ▶ By induction on the size of A .
- ▶ If A is a singleton (cardinality of A equal to 1), then, by reflexivity of $\succsim$, A 's only element is minimal.
- ▶ For the inductive step, let A be of cardinality $n + 1$ (where $n \geq 1$) and take any $x \in A$.
 - ▶ The set $A - \{x\}$ is of cardinality n and by the inductive assumption has a minimal element which we denote by y .
 - ▶ By completeness of $\succsim$, one of the two following will be satisfied:
 - ▶ If $x \succsim y$, then y is minimal in A .
 - ▶ If $y \succsim x$, then by transitivity $z \succsim x$ (because $z \succsim y$) for all $z \in A - \{x\}$, and thus x is minimal.



Proposition 4

If $\succsim$ is a rational preference relation on a finite set X , then $\succsim$ has a utility representation with values that are natural numbers.

(mapping a finite set in an order-preserving way onto a countably infinite set!)

Proof:

1. construct a sequence of "indifference" sets inductively:

- ▶ Let X_1 be the set of elements that are minimal in X .
- ▶ Assume we have constructed the non-empty sets $X_1, \dots, X_k$.
- ▶ If $X = X_1 \cup X_2 \cup \dots \cup X_k$, we are done.
- ▶ If not, define X_{k+1} to be the set of minimal elements in $X - X_1 - X_2 - \dots - X_k$. By the lemma, $X_{k+1} \neq \emptyset$. Because X is finite, we must be done after at most $|X|$ steps.
- ▶ Note that set X_k contains all elements which are at rank k in the preference relation (with 1 being the rank of least preferred alternatives in X).

2. define a utility function: let $u(x) = k$ if $x \in X_k$.
3. verify that u represents $\succsim$: for any $a, b \in X$,
 $a \succ b \Leftrightarrow a \notin X_1 \cup X_2 \cup \dots \cup X_{u(b)} \Leftrightarrow u(a) > u(b)$, and
 $a \sim b \Leftrightarrow a, b \in X_k \Leftrightarrow u(a) = u(b) = k$.



Existence of utility representations extends to all countable sets X

Every rational preference relation over a countable set X has a utility representation.

A set is countable if all its members can be enumerated, $\{x_n\}_{n=1,2,\dots}$, that is, we can count all its members with the positive integers.

A countable set can be finite or infinite (“you cannot finish counting”).

Proposition 5

If X is countable, then any rational preference relation $\succsim$ has a utility representation with a range $(-1, 1)$.

(mapping a countable set to an uncountable set (higher cardinality) in an order-preserving way!)

Proof of Proposition 5

- ▶ Let $\{x_n\}_{n=1,2,\dots}$ be an enumeration of all elements in X .
- ▶ Construct a utility function inductively.
 - ▶ Set $u(x_1) = 0$.
 - ▶ Assume that you have completed the definition of the values $u(x_1), \dots, u(x_{n-1})$ (all values in $(-1, 1)$) so that $x_k \succsim x_l$ iff $u(x_k) \geq u(x_l)$,
i.e. u represents the preference relation over $x_1, \dots, x_{n-1}$.
 - ▶ Now we define $u(x_n)$ (where $n \geq 2$).
 - ▶ If x_n is indifferent to x_k for some $k < n$, then assign $u(x_n) = u(x_k)$.
If not, then choose $u(x_n)$ to be $>$ than the maximum value of the set $\{u(x_k) | x_n \succ x_k\} \cup \{-1\}$ and $<$ than the minimum value of the set $\{u(x_k) | x_k \succ x_n\} \cup \{1\}$. This is possible since by transitivity all numbers in the first set are strictly below all numbers in the second set.
 - ▶ Thus, for any $k < n$ we have $x_n \succsim x_k$ iff $u(x_n) \geq u(x_k)$ and the function u extended to $\{x_1, \dots, x_n\}$ represents the preferences over those elements.

- ▶ To complete the proof that u represents $\succsim$, take any $x, y \in X$. For some k and l we have $x = x_k$ and $y = x_l$. Considering $n = \max\{k, l\}$ in the above implies that $x_k \succsim x_l$ iff $u(x_k) \geq u(x_l)$. □